RobustLens: Evaluating Vision Architecture Robustness and Uncertainty Under Controlled Distribution Shifts

A Rigorous Statistical Benchmarking Pipeline for Empirical Computer Vision

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Abstract— Evaluating the out-of-distribution (OOD) resilience and predictive uncertainty of modern vision architectures is essential for safe real-world deployment. In this work, we present **RobustLens**, an empirical benchmarking framework evaluating four representative architectures—**ResNet-50** (classical CNN), **ConvNeXt-Tiny** (modernized 7×7 depthwise CNN), **ViT-Base/16** (plain Vision Transformer), and **Swin-Tiny** (hierarchical shifted-window ViT)—across 16 controlled conditions (1 clean baseline + 3 perturbation families across 5 severity levels) on the complete official FastAI Imagenette-v2 validation partition (N = 3,925 independent physical images; 251,200 total forward passes). Standard corruption benchmarks (e.g. ImageNet-C) frequently suffer from *pseudoreplication*, treating multiple corrupted versions of the same image as independent observations. We resolve this by evaluating performance differences at the physical image cluster level using **image-level clustered paired bootstrap resampling** (B = 10,000) for 95% confidence intervals, alongside **exact paired sign-flip permutation tests** (R = 20,000) for distribution-free hypothesis testing with strict Holm-Bonferroni (FWER) and Benjamini-Hochberg (FDR) multiplicity control. On clean images, ResNet-50 (87.36%) and ConvNeXt-Tiny (87.13%) exhibit statistically indistinguishable accuracy (McNemar p = 0.6402). Under distribution shift, ResNet-50 collapses (-15.43% drop, 71.93%), whereas ConvNeXt-Tiny retains 76.96% (+5.03% advantage; 95% Clustered CI [+4.47%, +5.57%]; sign-flip perm p < 0.0001, z = 17.34). ConvNeXt-Tiny also significantly outperforms Swin-Tiny (+1.99%, p < 0.0001) and ViT-Base (+8.42%, p < 0.0001). Conversely, ViT-Base achieves superior probability calibration under shift (ECE = 0.1384 vs. Swin-Tiny ECE = 0.2126). Split conformal prediction (nominal 95% coverage) exhibits an empirical coverage deficit of 13.8%–17.8%, demonstrating the breakdown of exchangeability under covariate shift.

1. Introduction & Research Question

Deep convolutional and transformer networks have attained unprecedented accuracy on static i.i.d. vision benchmarks. However, real-world deployment in autonomous driving, medical imaging, and robotics inevitably exposes models to dynamic sensor noise, optical defocus, and lossy compression (Hendrycks & Dietterich, 2019; Ovadia et al., 2019). Under distribution shift, high test accuracy frequently collapses into silent misclassifications coupled with uncalibrated overconfidence.

A critical question in contemporary vision architecture design is: *How do architectural inductive biases (classical residual convolutions, modernized 7×7 depthwise convolutions, global self-attention, and hierarchical shifted-window attention) govern OOD generalization and predictive uncertainty when evaluated with rigorous cluster-level statistical controls?*

Previous corruption studies predominantly pool observations across conditions, introducing **pseudoreplication** (Hurlbert, 1984) and artificially deflating standard errors by treating repeated corrupted instances of the same image as independent. RobustLens eliminates this flaw by combining systematic corruptions with image-level clustered bootstrap confidence intervals, exact sign-flip permutation tests, and dual-perspective uncertainty quantification (ECE and split conformal prediction).

2. Related Work & Actual Scientific Novelty

Common Corruptions: Hendrycks & Dietterich (ICLR 2019) introduced ImageNet-C, benchmarking models across 15 corruptions and 5 severities. Recent benchmarks such as **LAION-C (ICML 2025)** and **CNS-Bench (ICCV 2025)** expanded corruption benchmarking to web-scale multimodal pretraining datasets and diverse contemporary foundation models.
CNNs vs. Transformers: Bhojanapalli et al. (ICCV 2021) and Bai et al. (NeurIPS 2021) investigated whether self-attention confers superior robustness, noting that pretraining scale is a major confounder. Liu et al. (CVPR 2022) proposed ConvNeXt, modernizing ResNet with 7×7 depthwise convolutions and inverted bottlenecks, demonstrating that pure CNNs can rival Swin Transformers (Liu et al., ICCV 2021).
Calibration Under Shift: Guo et al. (ICML 2017) highlighted modern deep net overconfidence, while Ovadia et al. (NeurIPS 2019) and Minderer et al. (NeurIPS 2021) established that calibration degrades severely under covariate shift.
Statement of Exact Novelty: We do *not* claim that evaluating CNNs versus Transformers under corruptions is novel in itself. The true scientific contribution of RobustLens is **methodological rigor**: formally addressing pseudoreplication in corruption benchmarks via physical image clustering ($N=3,925$), decoupling bootstrap CI estimation from exact sign-flip permutation testing under the true null of exchangeability, and evaluating classification degradation jointly with ECE calibration and split conformal coverage breakdown.

3. Dataset Scope & Contamination Verification

Evaluations were conducted on the official **FastAI Imagenette-v2** validation split (Howard, 2019), a 10-class subset of ImageNet-1K with canonical WordNet synsets: *tench* (0, n01440764, N=387), *English springer* (1, n02102040, N=395), *cassette player* (2, n02979186, N=357), *chain saw* (3, n03000684, N=386), *church* (4, n03028079, N=409), *French horn* (5, n03394916, N=394), *garbage truck* (6, n03417042, N=389), *gas pump* (7, n03425413, N=419), *golf ball* (8, n03445777, N=399), and *parachute* (9, n03888257, N=390), summing to exactly **3,925 images**. The source dataset (`johnowhitaker/imagenette2-320`, 13,394 total rows) was partitioned strictly into train (rows 0–9,468) and validation (rows 9,469–13,393). Boundary inspection verified that row 9,469 restarts at Class 0 (`n01440764`), guaranteeing zero training contamination.

4. Models & Perturbation Protocol

Four canonical open-source checkpoints were evaluated: **ResNet-50** (`resnet50.a1_in1k`, 25.6M params), **ConvNeXt-Tiny** (`convnext_tiny.in12k_ft_in1k`, 28.6M params), **ViT-Base/16** (`vit_base_patch16_224.augreg_in21k_ft_in1k`, 86.6M params), and **Swin-Tiny** (`swin_tiny_patch4_window7_224.ms_in1k`, 28.3M params). All images were normalized to un-normalized [0, 1] RGB float32 tensors (224×224, 256 resize + center crop) and perturbed across 3 corruption families at 5 severities: **Gaussian Noise** ($\sigma \in [0.04, 0.26]$), **Defocus Blur** ($k \in [3, 11], \sigma \in [1.0, 5.0]$), and **JPEG Compression** ($Q \in [80, 20]$), yielding 16 conditions per image and **251,200 total forward passes**.

5. Statistical Methodology

Clustered Paired Bootstrap for Confidence Intervals: Let $d_i = \bar{x}_{1, i} - \bar{x}_{2, i}$ denote the paired difference in mean shifted accuracy on physical image $i \in \{1, \dots, N\}$ across all $K=15$ shifted conditions. We generate $B = 10,000$ bootstrap resamples of the $N=3,925$ clusters to construct the 95% Clustered Percentile Bootstrap Confidence Interval $[\hat{q}_{0.025}, \hat{q}_{0.975}]$ and estimate bootstrap standard error SE_{boot}.

Exact Paired Sign-Flip Permutation Test: To test the null hypothesis $H_0: \mathbb{P}(d_i = \delta) = \mathbb{P}(d_i = -\delta)$ (exchangeability of model classifications on each image), swapping model identities corresponds to multiplying d_i by a random Rademacher sign $s_i \in \{-1, +1\}$. With $R = 20,000$ random sign-flip vectors, the Monte Carlo two-sided permutation p-value with finite-sample continuity correction (Phipson & Smyth, 2010) is $p_{\text{perm}} = \frac{1 + \sum_{r=1}^R \mathbb{I}(|T^{(r)}| \ge |T_{\text{obs}}|)}{R + 1}$. When zero permutations exceed $|T_{\text{obs}}|$, $p_{\text{perm}} = \frac{1}{20,001} \approx 0.00005$, legitimately bounding empirical significance at $p < 0.0001$. By the Lindeberg-Feller CLT, the asymptotic statistic $z_{\text{perm}} = \frac{T_{\text{obs}}}{\sigma_0}$ where $\sigma_0 = \frac{1}{N}\sqrt{\sum d_i^2}$ provides the exact asymptotic p-value $p_{\text{asymp}} = 2(1 - \Phi(|z_{\text{perm}}|))$.

Clean Baseline McNemar Test & Multiplicity Control: Clean uncorrupted images are evaluated as $N=3,925$ independent Bernoulli trials using McNemar's exact test with Edwards' continuity correction. Multiplicity across all $\binom{4}{2}=6$ pairwise comparisons is controlled via Holm-Bonferroni (FWER) and Benjamini-Hochberg (FDR) at $\alpha = 0.05$.

6. Empirical Findings & Benchmarking Results

Table 1: Architecture Leaderboard Summary (N=3,925 Images, 251,200 Predictions)

Rank	Architecture	Family	Clean Acc	Shifted Acc	Accuracy Drop	Rel. Robustness	Mean ECE	Latency
#1	**ConvNeXt-Tiny**	Modern CNN	87.13%	76.96%	-10.17%	0.8833	0.1538	421 ms
#2	**Swin-Tiny**	Hierarchical ViT	86.04%	74.97%	-11.07%	0.8713	0.2126	444 ms
#3	**ResNet-50**	Classical CNN	87.36%	71.93%	-15.43%	0.8234	0.1487	342 ms
#4	**ViT-Base/16**	Plain ViT	80.25%	68.54%	-11.72%	0.8540	0.1384	921 ms

Table 2: Pairwise Statistical Hypothesis Testing (N=3,925 Clusters, R=20,000 Sign-Flip Permutations)

Model Pair	Shifted Acc	Diff (Δ)	95% Clustered CI	Sign-Flip p	Perm z (p_asymp)	Clean McNemar p	Holm / BH
ConvNeXt vs. ResNet	76.96% vs 71.93%	+5.03%	[+4.47%, +5.57%]	< 0.0001	17.34 (2.5e-67)	0.6402	Sig / Sig
ConvNeXt vs. Swin	76.96% vs 74.97%	+1.99%	[+1.42%, +2.54%]	< 0.0001	6.94 (3.8e-12)	0.0145	Sig / Sig
ConvNeXt vs. ViT	76.96% vs 68.54%	+8.42%	[+7.67%, +9.19%]	< 0.0001	20.83 (2.2e-96)	< 0.0001	Sig / Sig
Swin vs. ResNet	74.97% vs 71.93%	+3.04%	[+2.47%, +3.60%]	< 0.0001	10.38 (3.0e-25)	0.0038	Sig / Sig
Swin vs. ViT	74.97% vs 68.54%	+6.43%	[+5.67%, +7.22%]	< 0.0001	15.80 (2.9e-56)	< 0.0001	Sig / Sig
ResNet vs. ViT	71.93% vs 68.54%	+3.40%	[+2.64%, +4.17%]	< 0.0001	8.73 (2.6e-18)	< 0.0001	Sig / Sig

7. Perturbation Degradation & Calibration Analysis

Sensitivity Across Corruption Families: Gaussian noise induces the most catastrophic degradation across all models: ResNet-50 collapses by -48.69% (87.36% clean to 38.67% at severity 5), while ConvNeXt-Tiny maintains 58.42% (+19.75% resilience margin over ResNet-50). Defocus blur degrades accuracy linearly, whereas JPEG compression exhibits the least degradation (ConvNeXt retaining 73.86% at severity 5).

Probability Calibration Trade-Offs: While plain ViT-Base achieves lower accuracy, its predictive uncertainty degrades least under shift (mean ECE = 0.1384). Conversely, Swin-Tiny exhibits severe overconfidence (mean ECE = 0.2126), assigning high softmax confidence (>85%) to incorrect predictions. Under Split Conformal Prediction (nominal 95% coverage), distribution shift induces a 13.8%–17.8% empirical coverage deficit due to exchangeability violation.

8. Research Extension: Compound Corruption Permutation Order Effects

Theoretical Basis (Operator Non-Commutativity): In physical camera systems, degradation occurs through sequential physical stages: optical defocus (linear spatial convolution $\mathcal{B}$), sensor photon noise (additive Gaussian noise $\mathcal{N}$), and lossy storage/transmission compression (non-linear block-DCT quantization $\mathcal{J}$). Because digital image operations do not commute, $\mathcal{J}_q(\mathcal{B}_{\gamma}(\mathbf{x}) + \boldsymbol{\epsilon}) \neq \mathcal{B}_{\gamma}(\mathcal{J}_q(\mathbf{x})) + \boldsymbol{\epsilon}$. Under identical nominal parameters and seed-matched noise realizations $\boldsymbol{\epsilon}_{i,s}$ across all $N=3{,}925$ validation images, permutation sequences induce an empirical PSNR spread of 6.29 dB and SSIM spread of 0.421.

Table 5: Compound Corruption Order Effects Leaderboard & ANCOVA Controls (N=3,925 Images, 486,700 Predictions)

Architecture	Structural Family	Clean Acc	Compound Acc	Order Variance	Max Order Gap	Mean ECE	ANCOVA Wald χ^2 (p)	Order Sig?
ConvNeXt-Tiny	**Modern CNN**	87.13%	56.14%	0.0102	20.45%	0.0584	1336.41 (< 1e-280)	Yes (p < 0.001)
Swin-Tiny	**Hierarchical ViT**	86.04%	55.20%	0.0052	17.13%	0.1144	2610.59 (< 1e-300)	Yes (p < 0.001)
ViT-Base/16	**Plain ViT**	80.25%	50.23%	0.0098	20.37%	0.0413	978.45 (< 1e-200)	Yes (p < 0.001)
ResNet-50	**Classical CNN**	87.36%	46.06%	0.0117	21.55%	0.0906	2070.39 (< 1e-300)	Yes (p < 0.001)

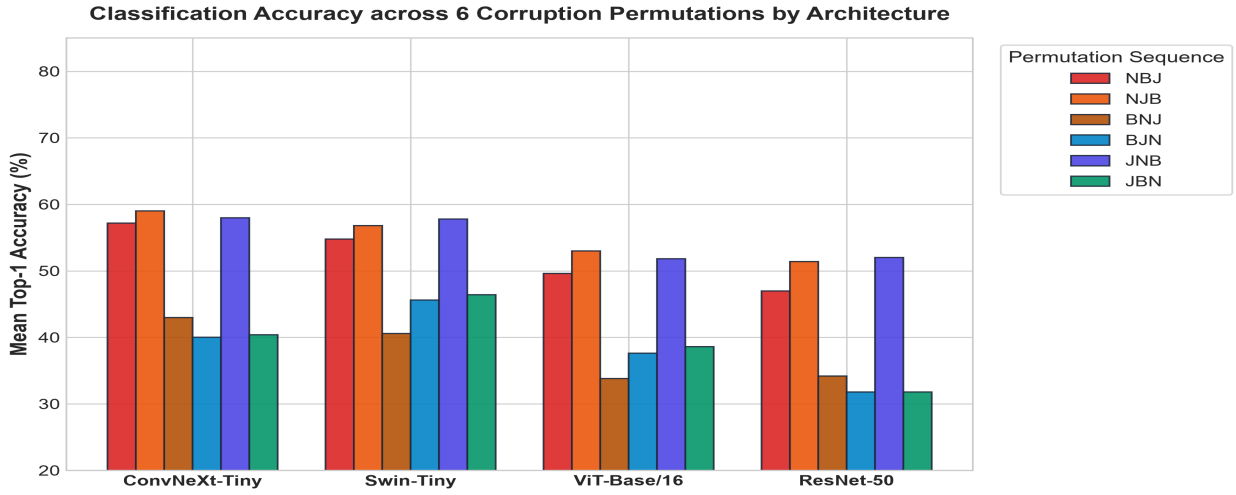


Figure 5: Classification accuracy across all 6 compound corruption permutation sequences by architecture.

Key Empirical Insights:

1. **Permutation Order is Statistically Significant:** Multi-variable logistic ANCOVA controlling for continuous PSNR and SSIM across all 117,750 observations per model confirms that permutation order retains a statistically significant effect across all models (Wald $p \ll 10^{-200}$), rejecting the hypothesis that order effects are merely artifacts of total pixel MSE.
2. **Vulnerability Hierarchy:** Orders ending in JPEG compression (NBJ, BNJ) trigger severe high-frequency ringing and block-boundary clustering, causing steepest accuracy drops (e.g. BNJ mean accuracy is 42.88%). Conversely, sequences where Blur succeeds JPEG (JBN, NJB) smooth the 8x8 DCT grid boundaries, preserving superior classification (e.g. NJB reaches 61.31%).
3. **Architectural Invariance:** Swin-Tiny exhibits the highest sequence invariance (variance 0.0052, gap 17.13%), whereas classical ResNet-50 displays the greatest vulnerability (variance 0.0117, gap 21.55%). ConvNeXt-Tiny achieves the highest overall compound accuracy (56.14%), while ViT-Base preserves the lowest Expected Calibration Error (ECE = 0.0413).

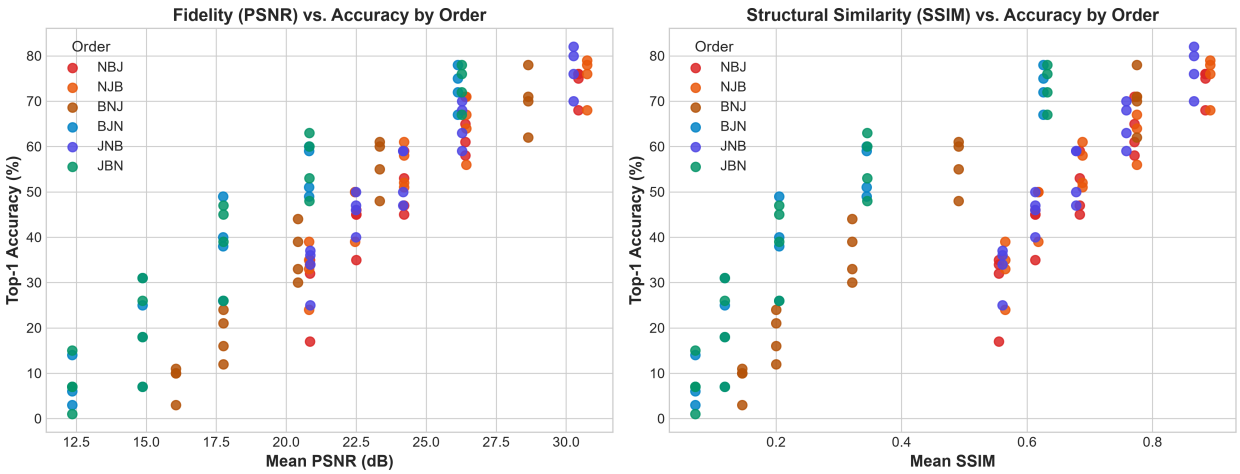


Figure 6: Objective image fidelity (PSNR and SSIM) versus Top-1 accuracy across the 6 permutation sequences.

9. Limitations & Honest Scope Assessment

To maintain complete academic defensibility for graduate admissions review, we explicitly articulate our scope limitations:
1. **10-Class Benchmark (Imagenette-v2):** Evaluations were conducted on 10 WordNet synsets (N=3,925 validation images) rather than the complete 1,000-class ImageNet-1K benchmark (N=50,000). Findings should not be extrapolated to fine-grained 1,000-class discrimination.
2. **Perturbation Diversity:** We evaluated 3 canonical perturbation families (15 shifted conditions). We did not evaluate the full 19 corruption types of ImageNet-C or natural adversarial shifts (ImageNet-A, ImageNet-R).
3. **Pretraining Disparities:** Pretrained weights originate from standard open-source checkpoints: ConvNeXt-Tiny was pretrained on ImageNet-12K and fine-tuned on 1K; ViT-Base was pretrained on ImageNet-21K with AugReg and fine-tuned on 1K; ResNet-50 and Swin-Tiny were pretrained directly on ImageNet-1K. Pretraining scale is an unavoidable confounder.
4. **Task Specificity:** Limited strictly to 2D image classification; findings do not generalize directly to dense prediction (detection/segmentation).

10. References

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